

Cirqlback Arcade — v2 Master Plan & Guide

The plan for **one unified pixel arcade** that houses everything Cirqlback plays: the 57 quick *circular* games and a new premium line of **ten truly-retro 16-bit cabinets** themed to local shops. One front door, one player identity, one look. This document is both the **vision/guide** and the **project plan**; a companion *Linear Plan* mirrors it as issues.

Decisions locked (2026-07-06). 16-bit is the house art tier · one unified "Cirqlback Arcade" front door · your custom avatar **stars as the playable hero in every cabinet** · the quarter is **free-play cinema** (never a paywall) · **all 10 cabinets open from day one** · the circular line gets a **light pixel reskin** so it sits natively in the pixel lobby.

Part 1 — The Vision

What we're building

A single **Cirqlback Arcade**: a pixel/CRT lobby (the look from the reference mockup) with a category rail, a player-profile card, and cards for every game. Inside live **two complementary lines**:

- **Circles** — the 57 existing one-thumb snacks (30–60s loops), lightly pixel-reskinned to match the lobby.
- **Main Street** — 10 new 16-bit cabinets: original homages to arcade classics, themed to local shops, using the **whole screen** so they go long (many levels, bosses, a high-score chase).

Both share one backend, one Daily system, one leaderboard/rewards spine, and **one customizable avatar**.

Two principles that never bend

- **Round is the mascot, not the cage.** Full rectangular arcade playfields — but every cabinet has a signature round hero tied to its shop (a mug, a pizza, a vinyl, a donut, a bubble), shaded like a lit sphere. The circle is the star, not the boundary.
- **Original always — homage, not clone.** Our own names, art, audio and themes throughout. We homage the *genre and mechanics* of the classics (fair game); we never reproduce trademarked names, characters, art, or sound.

Every cabinet must clear this bar

- ☐ **Visually exciting** — 16-bit shading, gradient skies, parallax, palette flashes.
 - ☐ **Many progressive, interesting levels** — themed worlds, a new mechanic per world, a boss cadence, a real difficulty ramp.
 - ☐ **Lots of juice & action** — screen shake, particles, hitstop, combo popups, satisfying SFX.
 - ☐ **Very playable** — tight controls, two-thumb where two actions overlap, instant restart.
 - ☐ **Funner / more interesting than the original** — one signature twist the classic never had.
 - ☐ **Great original music** — a unique 16-bit theme (+ boss variant + game-over jingle).
 - ☐ **A daily challenge** — a fixed daily seed everyone competes on (Freestyle is retired for this line).
 - ☐ **Your avatar stars** — the playable hero is your customized toy-person, re-skinned for the shop.
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Part 2 — The Foundation (the platform layer)

Built once in Phase 0, shared by all 10 cabinets and the lobby.

RetroEngine (16-bit tier)

The shared engine every cabinet extends: an internal **pixel buffer + nearest-neighbor upscale**, a **full-color draw kit** with `shade()` / `mix()` / `vgrad()` gradients, `shelf()` 3-tone surfaces, and `ball()` spherically-lit round heroes (all prototyped in the lookbooks). Bundled **3×5 pixel font** (CSP-safe, no external font). A toggleable, cheap **CRT** post-process (scanlines + vignette). Fixed-timestep update, **d-pad + tap + two-thumb** input, pause/menu/game-over chrome. Reuses the existing `gameId`-keyed backend for best score + Daily.

MusicKit

A layered chiptune/16-bit kit extending the current synth: square lead + harmony, triangle bass, a noise-drum channel. Each cabinet gets a **theme + boss/intense variant + game-over jingle**, with **dynamic layering** (music thickens as combo/level rises). A shared **Main Street leitmotif** each cabinet riffs on for sonic cohesion. Respects the existing mute toggle.

Insert-Coin cutscene

On launch, a reusable 16-bit sequence: your toy hand drops a quarter → the coin-slot lights → the CRT boots → the cabinet marquee → **PRESS START. Skippable after first view.** Pure flavor — always free. Bonus: **attract/demo mode** on the lobby (cabinets auto-play a short demo like a real arcade).

Avatar system + customizer

A "**little people & toys**" avatar — chunky, rounded, collectible. Customization axes: body/base color, head (skin, eyes, expression), hair/hat, outfit, accessory, and an optional sidekick toy. The avatar appears in the **lobby**, the **cutscene** (your hand, your figure climbing in), **leaderboards**, and — the key move — as the **playable hero in every cabinet**, re-skinned per shop (barista apron, delivery helmet, tool belt). **Shop outfits unlock by playing that cabinet**, pulling players across the arcade. Unlocks flow from the platform reward economy (points/streaks/badges) — **cosmetic only, never power.**

Unified pixel lobby

Rebuild the front door as the pixel chooser: a **category rail** (Action / Puzzle / Arcade / Classic / Reflex / Strategy / Zen) spanning **both** lines, a **player-profile card** (avatar · level · ★ · badges), game cards with cover art + difficulty + Daily flag, and a **bottom nav** (Home / Leaderboard / Achievements / Favorites). Circular games get a light pixel reskin to sit natively.

Leaderboards & rewards spine

- **Leaderboards:** Daily + all-time per game, keyed to avatar + name; friends/territory later.
- **Reward events:** first play, daily streak, level/world clears, boss kills, personal bests → **platform points** buying **avatar cosmetics + real-shop perks** (the moat). Assisted/perked runs stay **off** the ranked board (consistent with what's already live).
- **Achievements/badges** per cabinet (this is the long-deferred Badge Phase 2 finally landing).
- **Anti-cheat:** the client engine is untrusted; the server validates submitted scores.

Mobile & accessibility

16-bit is landscape; phones are portrait — so a **rotate prompt / portrait-fit + on-screen controls** per cabinet, honoring the **two-thumb** rule (move *and* act at once). Colorblind-safe palettes, **reduced-motion** (CRT/shake off — engine already reads the setting), assist difficulty, and a CRT toggle for low-end devices.

Part 3 — The 10 Cabinets

#	Cabinet	Shop	Homage	Round hero	Signature twist
1	Cuppa Rush	Coffee	Tapper	Mugs / beans	Mugs are physics objects — bank-shots off the bumper; boss "regulars" order combos
2	Slice Route	Pizza	Paperboy	Pizzas / wheels	Branching streets — pick your route at forks for risk/reward; weather worlds
3	Rummage	Thrift	Pac-Man	A vintage button	The rack-maze reshuffles as you clear it; outfit power-ups (speed, disguise, magnet)
4	Dozen	Donut	Q*bert	The donut	Color-mix glazes — multi-hop to a target flavor; moving/conveyor crates
5	Fresh Batch	Bakery	Donkey Kong	Pie-tins / bagels	Grab-and-throw rolling pins; destructible obstacles; risky top-shelf bonuses
6	Sundae Stack	Ice-cream	Tetris	The scoops	Flavor-match combos on top of line-clears; per-level "build this sundae" goals
7	Spin City	Record	Marble Madness	The vinyl	The record <i>plays the level's track</i> — tempo is your time pressure; groove rails
8	Taco Stack	Taqueria	BurgerTime	Tomatoes / lime	Lime is weapon and combo builder; a spicy meter; order tickets set build priority
9	Fix-It	Hardware	Fix-It Felix Jr.	Bolts / wrecking ball	The wrecker adapts to your patterns; power-tool upgrades; renovation bonus rounds
10	Spin Cycle	Laundromat	Puzzle Bobble	Soap bubbles	The drum rotates the bubble grid; suds hazards; a bleach color-bomb

Each cabinet targets ~20–40 stages across 3 themed worlds, a boss every ~6–8 levels, and a new mechanic introduced per world. Endless not required; the **Daily** is a fixed daily seed.

Part 4 — Definition of Done (per cabinet)

Every cabinet ships only when all of this is true:

- ☐ Engine extends **RetroEngine (16-bit)**; round hero rendered with `ball()` shading.
 - ☐ **3 worlds, ~20–40 progressive levels**, a boss cadence, new-mechanic-per-world.
 - ☐ The **signature twist** is in and fun.
 - ☐ **Juice pass** — shake, particles, hitstop, combo popups, transitions.
 - ☐ **Your avatar** is the playable hero, with the shop outfit + its unlock wired.
 - ☐ **Original music** — theme + boss variant + game-over jingle, dynamic layering.
 - ☐ **Insert-Coin cutscene** wired on launch (skippable after first view).
 - ☐ **Daily** challenge live (fixed seed) + leaderboard submit + reward events.
 - ☐ **Two-thumb mobile controls** + portrait handling + reduced-motion respected.
 - ☐ `tsc` clean, verified in-browser (0 console errors), lazy-routed in the lobby.
 - ☐ **Plan + Linear updated, and pushed to Render** (auto-deploy) — done every cabinet.
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Part 5 — Project Plan

Phase 0 · Foundation

Build once; everything depends on it. The avatar rig is a hard dependency for cabinets (the hero is your avatar), so it lands before cabinet #1.

- ☑ **RetroEngine (16-bit kit + CRT)** — pixel buffer + nearest-neighbor upscale, shaded draw kit (`shade` / `mix` / `vgrad` / `shelf` / `ball`), bundled pixel font, CRT (scanlines + vignette), fixed-timestep loop, d-pad/tap/two-thumb input, chiptune primitives. *Shipped:* `client/src/game/retro-engine.ts` + `retro-font.ts` ; proven live at `/retro-demo` .
- ☑ **MusicKit** — layered chiptune (lead/harmony/bass/noise-drums), lookahead scheduler, dynamic layering by intensity, one-shot jingles, shared Main Street leitmotif. *Shipped:* `client/src/game/musickit.ts` ; proven live in `/retro-demo` (theme plays on first tap, thickens over time).
- ☑ **Avatar system + customizer** — toy figure, customization axes, per-shop outfit rig, localStorage + profile persistence. *Shipped:* `client/src/game/avatar.ts` + `/avatar` customizer; `RetroEngine.avatar()` draws the in-game hero. (Reward-driven unlock economy + lobby/leaderboard placement land with those pieces.)
- ☐ **Insert-Coin cutscene** — reusable, skippable, attract/demo mode.
- ☐ **Unified pixel lobby** — category rail, player-profile card, game cards, bottom nav; circular-line pixel reskin.
- ☐ **Leaderboards & rewards wiring** — Daily/all-time, reward events, server-side score validation.
- ☐ **Cuppa Rush POC** — the first cabinet, proving the entire stack end-to-end (**M0**).

Phase 1 · The 10 cabinets

Built in batches, each meeting the full Definition of Done. **After every cabinet: update the plan + Linear and push to Render.**
Suggested order (recommended-first three prove the range early):

1. **Cuppa Rush** (POC) · 2. **Slice Route** · 3. **Spin Cycle** · 4. **Rummage** · 5. **Dozen** · 6. **Sundae Stack** · 7. **Fresh Batch** · 8. **Taco Stack** · 9. **Spin City** · 10. **Fix-It**.

Phase 2 · Meta & polish

- ☐ Achievements/badges catalog (auto thresholds) per cabinet.
- ☐ Avatar cosmetic-unlock catalog + **reward "skin packs"** a business can hand out.
- ☐ Sponsorship/skin hooks — a real shop adopts its matching cabinet + attaches a perk.
- ☐ Social — share your score/avatar card; race the Daily #1 "ghost."
- ☐ Seasonal / limited-time skins + a rotating "cabinet of the week."
- ☐ Full accessibility & performance pass across all cabinets.

Milestones

- ☐ **M0** — Foundation done + Cuppa Rush playable end-to-end (engine, music, cutscene, avatar-hero, Daily, lobby entry).
 - ☐ **M1** — First 3 cabinets live (Cuppa Rush, Slice Route, Spin Cycle) + lobby unified.
 - ☐ **M2** — All 10 cabinets live.
 - ☐ **M3** — Meta & polish complete (achievements, sponsorship, social, seasonal, a11y).
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Part 6 — Risks & Notes

- **Scope is real** — each cabinet is a full level-based arcade game with music and avatar art; batches of a few, not a sweep. Procedural generation + hand-tuned seeds keep level content scalable.
 - **Foundation-heavy Phase 0** — the avatar rig, MusicKit, cutscene, and unified lobby all precede cabinet #1. This is deliberate: it makes every later cabinet cheap and consistent.
 - **CRT + juice on mobile** — keep the post-process cheap; everything toggleable.
 - **Fairness** — rewards are cosmetic/perk only; assisted runs stay off ranked boards; the server validates scores.
 - **Stay original** — original names/art/audio/themes throughout; homage the mechanics, never the trademarked assets.
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Ready to build: Phase 0 (RetroEngine 16-bit + MusicKit + Avatar system + Insert-Coin cutscene + unified lobby + Cuppa Rush POC) is milestone M0.